

Turf Booking Manager

Project Report

VITyarthi - Build Your Own Project

1. Introduction

Turf Booking Manager is a Python console application designed to make sports turf booking easier. It allows users to check availability, create bookings, search and cancel bookings, and view simple reports.

2. Problem Statement

Manual turf booking can make it difficult to track occupied slots and customer information. The project provides a simple menu-driven system for handling these tasks.

3. Functional Requirements

The system provides availability checking, slot booking, viewing bookings, searching bookings, cancelling bookings, viewing recent bookings and displaying summary information.

4. Non-functional Requirements

Usability: simple numbered menu. Reliability: validation of common user inputs. Maintainability: functions and separate Python modules. Error handling: invalid dates, numbers and phone numbers are handled.

5. System Architecture

The application is divided into data, validation, booking, display, reporting and main modules. The main program connects the modules while the data module stores the current bookings.

6. Design Diagrams

The accompanying documentation folder contains the architecture, workflow, use case, component/class and sequence diagrams.

7. Design Decisions and Rationale

Python lists and dictionaries were selected because they are suitable for a beginner-level console application. The program is split into small modules so each part has a clear purpose.

8. Implementation Details

The program uses functions, loops, conditions, lists, dictionaries, string methods, random numbers and datetime validation. Bookings remain in memory during execution.

9. Screenshots / Results

Screenshots can be added after running the program. The main outputs include the menu, availability display, booking confirmation and summary.

10. Testing Approach

A simple test file checks creation of a booking, detection of an occupied slot and cancellation of a booking.

11. Challenges Faced

Important challenges include validating user input, preventing duplicate booking IDs and preventing a slot from being booked twice.

12. Learnings and Key Takeaways

The project demonstrates how Python functions, lists, dictionaries, loops and conditions can be combined to build a complete console application.

13. Future Enhancements

Future versions could add permanent database storage, staff login, online payment, a graphical interface and more detailed reports.

14. References

Course/project instructions provided by VITYarthi. Python standard library documentation can be consulted for datetime and random.

Functional Module Summary

Module	Purpose
Availability	Shows free and booked slots
Booking	Creates and cancels bookings
Validation	Checks dates, numbers and phone numbers
Reports	Searches bookings and displays summaries
Display	Shows menus and booking information